

Course Code: CS-307	Course Name: Computer Networks
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Student Roll No:	Section No:

Instructions:

- Return the question paper.
- **All questions must be answered in answer script and according to the sequence given in the question paper.**
- Each question carry 10 marks.
- In case of any ambiguity, you may make an assumption, however, your assumption should not contradict any statement in the question paper.

Time: 60 minutes.

Max Marks: 60

Question 1: Design a reliable data transfer protocol between the sender and receiver, operating over a channel that can lose or corrupt packets (data or ACKs) and error-prone. Specify the protocol using finite state machine for the sender side.

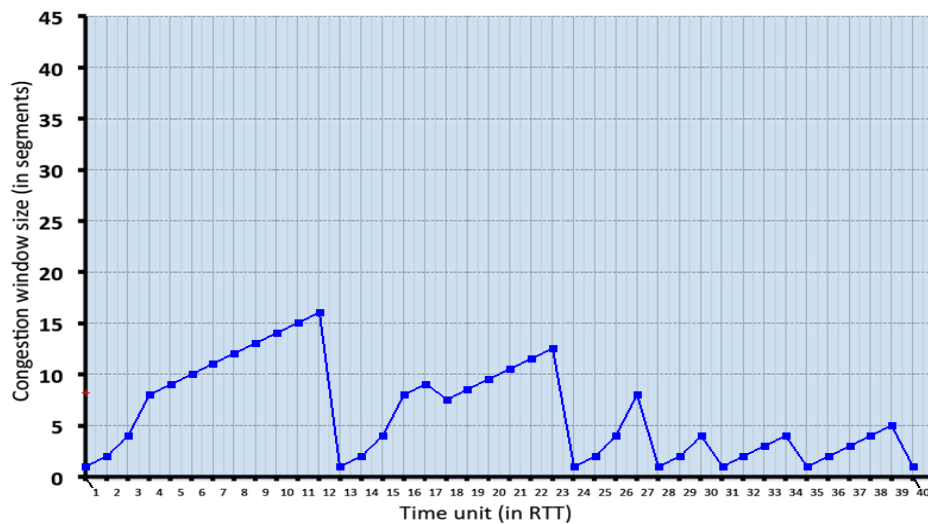
Question 2: Suppose a process in Host A has a UDP socket with port number 1234. Suppose both Host A and Host B each sends a UDP segment to Host C with destination port number 1234.

- Will both of these segments be directed to the same socket at Host C? Explain.
- In this case, how will the process at Host C know that these two segments are originated from two different hosts?

Question 3: For a p2p file-sharing application, do you agree with the statement, "There is no notion of client and server sides of a communication session"? Why or why not?

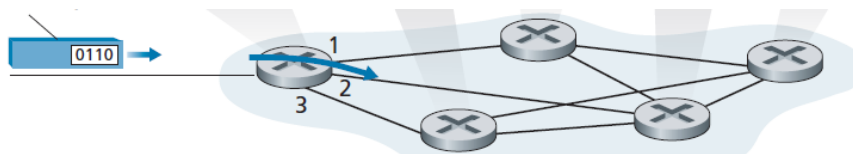
Question 4: Draw a timeline diagram for the Go-Back-N algorithm with the sender's window sizes equal to 4. Assume at $t=0$, the sender sends packets 0, 1, 2, 3, 4, and packet 2 will be lost.

Question 5: Consider the figure below, which plots the evolution of TCP's congestion window at the beginning of each time unit (where the unit of time is equal to the RTT); **see Figure 3.53 in the text**. In the abstract model for this problem, TCP sends a "flight" of packets of size $cwnd$ at the beginning of each time unit. The result of sending that flight of packets is that either (i) all packets are ACKed at the end of the time unit, (ii) there is a timeout for the first packet, or (iii) there is a triple duplicate ACK for the first packet. In this problem, you are asked to reconstruct the sequence of events (ACKs, losses) that resulted in the evolution of TCP's $cwnd$ shown below.



- Give the times at which TCP is in slow start.
- Give the times at which TCP is in congestion avoidance.
- Give the times at which TCP is in fast recovery.
- Give the times at which packets are lost via timeout.
- Give the times at which packets are lost via *triple ACK*.
- Give the times at which the value of *ssthresh* changes (if it changes between $t=3$ and $t=4$, use $t=4$ in your answer)

Question 6: Consider a group of routers in the core network as shown in the figure below.



- Show in steps, how data packets entering one interface of a router are forwarded to an output interface of the same router.
- How routing helps forwarding? Explain.

BEST OF LUCK!